

Technical change and health: Evidence from the reduction of sugarcane burning in Brazil*

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Abstract

São Paulo, the major sugarcane-producing state in Brazil, commands over a fifth of the world's sugarcane production. This paper examines how the transition from manual pre-harvest burning to mechanical harvesting affected public health across São Paulo municipalities. Using a difference-in-differences design that exploits variation in municipal sugarcane land share, we find that greater exposure to the technical change reduced hospitalizations for respiratory and circulatory diseases. Most distinctively, we document reductions in pollution-linked congenital anomalies and in respiratory mortality among children under five. Adult mortality from respiratory and circulatory causes shows no effect. We directly confirm the smoke mechanism: fire incidence on sugarcane fields and particulate matter concentrations fell exclusively during the harvest season, and hospitalization reductions were disproportionately concentrated in those same months. Robustness is corroborated by wind direction: municipalities downwind of sugarcane areas benefited more. Placebo tests across other diagnostic groups confirm that effects concentrate precisely where the biological mechanism predicts.

Keywords: Sugarcane Burning; Technical Change; Health Outcomes; Air Pollution.

JEL Codes: Q52, Q55, I18.

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1 Introduction

Agricultural fires have several drawbacks during and after burning, including alterations to soil characteristics and significant local air pollution ([Andreae and Merlet, 2001](#); [Santín and Doerr, 2016](#)). Despite these disadvantages, fires remain widely used in agriculture because they reduce costs associated with clearing bushes, crop residue management, and controlling invasive plant species ([Levine, 1991](#); [Radosevich et al., 2007](#)). This cost reduction is particularly important for smallholders, but fires are also common in large-scale agriculture – *e.g.*, pre-harvest sugarcane burning in Brazil ([Uriarte et al., 2009](#)) and wheat post-harvest in China, India, Pakistan, Nepal, and Bangladesh ([Lin and Begho, 2022](#); [Li et al., 2008](#)).

Brazil dominates the world sugarcane market, and São Paulo state sits at its center – alone accounting for over a fifth of global production.¹ The state underwent a significant shift in harvesting technology. In 2007, roughly one third of sugarcane was harvested mechanically; by 2015, that share had risen to over 95% ([Conab, 2024](#)). As a consequence, the use of pre-harvest burning – and the air pollution it generates – declined sharply across the state. The goal of this paper is to examine how this shift from manual burning-based harvesting to mechanical harvesting affected public health.

We exploit the uneven distribution of sugarcane production across São Paulo municipalities to quantify exposure to the technical change. While half of all municipalities allocate minimal or no land to sugarcane cultivation, the other half dedicates at least 5% of their territory to this crop, and the top decile allocates over 45% ([IBGE, 2024](#)). Individuals in municipalities with more land allocated to sugarcane were more exposed to sugarcane smoke before the transition and stand to benefit more from its elimination. This setting is well suited for a difference-in-differences design with continuous treatment, where the state-level timing of the burning reduction interacts with the municipality-level sugarcane land share to generate variation in exposure to the technical change that is plausibly exogenous to local health trends ([Callaway and Sant’Anna, 2021](#)).

We combine several data sources. Our primary health outcomes come from three Ministry of Health registries: hospital admissions for respiratory (ICD-10 J00–J99) and cardiovascular conditions (ICD-10 I00–I99) from the Hospital Information System (SIH/SUS) over 2008–2022; birth records – including weight, gestational age, detailed anomaly information, and maternal characteristics – from the Live Birth Information System (Sinasc)

¹Brazil accounted for 37% of global sugarcane production in 2022. São Paulo, as Brazil’s primary producer, contributes 58% to the nation’s output, representing approximately 21% of world production ([FAO, 2025](#)).

over 2007–2023; and deaths attributed to respiratory and cardiovascular causes from the Mortality Information System (SIM) over 2002–2023. Our treatment variable is the pre-2011 average share of municipal land area dedicated to sugarcane, constructed from IBGE’s Municipal Agricultural Survey (PAM). To document the pollution channel, we use daily PM_{10} and $PM_{2.5}$ readings from Companhia Ambiental do Estado de São Paulo (Cetesb) monitoring stations, and verify fire incidence by cross-referencing satellite burned-area data (Modis) with sugarcane land cover (MapBiomas). To test whether health improvements are larger in municipalities exposed to smoke carried by prevailing winds, we construct a downwind exposure measure from Nasa Power (Merra-2) wind direction for the harvest months.

We find that the transition to mechanical harvesting reduced hospitalizations for both respiratory and circulatory diseases: a one-standard-deviation increase in exposure ($SD = 0.19$) reduces respiratory hospitalizations by 7.2% and circulatory hospitalizations by 4.5%. Birthweight outcomes show no effect, suggesting that the pollution reductions lie below the threshold for general fetal growth restriction. We do, however, document reductions in congenital anomalies – and crucially, the effect is concentrated among anomaly types with established biological pathways to particulate matter exposure. A one-standard-deviation increase in exposure reduces the incidence of the pollution-linked anomaly index – comprising neural tube defects, orofacial clefts, and cardiac malformations – by 22.6% relative to baseline, driven primarily by cardiac malformations (–38.3%) and neural tube defects (–9.4%). Adult mortality rates from respiratory and circulatory causes show no change. In contrast, respiratory mortality among children under five is reduced by approximately 11% at one standard deviation of exposure, indicating that the pollution reductions are large enough to shift mortality at the most vulnerable margin.

These health improvements operate through the smoke channel, which we document directly using two independent measures. First, mechanical harvesting reduced PM_{10} and $PM_{2.5}$ concentrations during the harvest season (July–September), with no detectable effect in the off-season. Second, fire incidence on sugarcane fields fell sharply during the harvest season, directly confirming that the PM reductions reflect less burning rather than secular improvements in air quality. The causal interpretation is further corroborated by complementary strategies. Municipalities downwind of sugarcane areas experienced larger health improvements, confirming that effects travel with prevailing winds during the harvest season. Placebo tests across all ICD-10 diagnostic groups yield null effects for diseases with no biological link to particulate matter, ruling out the possibility that our estimates reflect broad socioeconomic trends in heavily sugarcane-dependent municipalities. A continuous

treatment measure replaces the binary post-2010 indicator with the observed fraction of sugarcane burning emissions already eliminated in each year, and yields effects consistent in sign and proportional in magnitude to the baseline estimates.

This evidence matters beyond São Paulo: several Brazilian states continue to rely on pre-harvest burning. In the Northeast, where only one-third of the sugarcane harvest is mechanized (Conab, 2024), the practice remains widespread. The use of fire in sugarcane cultivation remains common around the world as well – including in Florida (Nowell et al., 2022)², India (Jethva et al., 2019), and Thailand (Phairuang et al., 2019; Prasertsri et al., 2025). Our findings thus speak to a broader tension in agricultural production: practices that reduce short-run costs but impose lasting health burdens on surrounding populations. For one of the world’s largest agricultural burning practices, we estimate these costs at R\$ 232 million (USD 59 million) in avoided hospitalizations and congenital anomalies over a decade – a conservative lower bound that excludes outpatient visits, productivity losses, and the lifetime social protection costs associated with severe birth defects. Quantifying these externalities is a necessary step toward designing policies that align private incentives with public health.

This paper contributes to the literature on the health effects of air pollution (Burnett et al., 2018; Turner et al., 2020) and to studies examining the health impacts of environmental regulations and agreements (Isen et al., 2017; McCarty et al., 2007). Prior research has linked air pollution exposure to adverse birth outcomes (Currie and Schmieder, 2009; Sram et al., 2013; Greenstone and Hanna, 2014; Van Rossem et al., 2015), respiratory illnesses (Mo et al., 2018; Simkovich et al., 2019; Tran et al., 2023), and circulatory diseases (Franchini and Mannucci, 2007; Lee et al., 2014; Franklin et al., 2015). We extend this literature by providing estimates of the health consequences of a large-scale reduction in agricultural burning – a source of pollution that has received limited attention relative to its scale. Previous research has investigated pre-harvest sugarcane burning in Brazil without identifying health effects at scale (Valente and Laurini, 2021; Cristale et al., 2012; Souto-Oliveira et al., 2023; França et al., 2014). Some studies examine the relationship between sugarcane smoke and health in specific municipalities (Arbex et al., 2000; Cançado et al., 2006) or among workers directly exposed (Prado et al., 2012; Abreu et al., 2011; Ribeiro and Ficarelli, 2010).

Rangel and Vogl (2019) and Chagas et al. (2016) are the closest antecedents to our

²Florida presents a striking case: pre-harvest burning persists despite the harvest being fully mechanized. Growers argue that burning reduces costs and accelerates the harvest by eliminating leaf trash before mechanical cutting. Critics counter that the practice continues primarily because of the political influence of the sugar industry at the state level (Nowell et al., 2022).

paper, both studying health outcomes in São Paulo. [Rangel and Vogl \(2019\)](#) examines the impact of agricultural fires on birth outcomes, exploiting daily variation in wind direction and fire location to isolate smoke exposure. Upwind fires led to lower birthweight and shorter gestational age, while non-upwind fires had only mild effects. [Chagas et al. \(2016\)](#) estimates the effects of sugarcane production on respiratory hospitalizations, introducing a spatial difference-in-differences design that accounts for spillover effects onto neighboring non-producing municipalities.

Our paper builds on these contributions along three dimensions. First, we broaden the set of outcomes. While [Chagas et al. \(2016\)](#) focuses on respiratory hospitalizations, we also examine circulatory hospitalizations, infant mortality, and pollution-linked congenital anomalies. Infant mortality has not been examined in this context, despite being a standard indicator of population vulnerability to environmental exposures. Cardiac malformations and neural tube defects have established biological pathways to particulate matter exposure ([Tanwar et al., 2017](#); [Chen et al., 2024](#)), yet have not been studied in the sugarcane burning literature. While [Rangel and Vogl \(2019\)](#) examines birthweight and gestational age, our anomaly index captures a margin of fetal damage that is both biologically grounded and previously undocumented. Second, our sample period is better suited to isolating the smoke channel. Our data span a sharp decline in pre-harvest burning while cultivated area remained broadly stable – a contrast that [Chagas et al. \(2016\)](#) could not exploit, as their sample runs through a period of rapid sugarcane expansion in which burning and production moved together. Third, our exposure measure better captures treatment intensity. [Chagas et al. \(2016\)](#) assigns municipalities to treated and control groups based on whether their sugarcane land share falls above or below the median (6.7% in their sample) – a reasonable choice given the state of the literature at the time, but one that conflates municipalities with vastly different exposure levels. A municipality allocating 6.8% of its land to sugarcane and one allocating 50% are assigned identical treatment status, while a municipality at 6.6% – nearly indistinguishable in land use – falls into the control group. Our continuous design instead weights each municipality by its pre-treatment sugarcane land share, so that identification is proportional to actual exposure intensity rather than the median threshold.

2 Background

2.1 Sugarcane production in Brazil

Brazil is the world's leading producer of sugarcane, harvesting approximately 760 million tonnes in 2024 – 40% of global output (Figure A-1) (FAO, 2025). Domestically, sugarcane is split roughly evenly between two uses: sugar and ethanol (Conab, 2024). This balance shifts at the margin in response to market conditions – *e.g.*, the sugar share rises when international prices are favorable to producers, and the ethanol share expands when policies that mandate or subsidize biofuel consumption are tightened, or when higher international oil prices make ethanol more competitive at the pump (Goldemberg et al., 2018).

Between 2000 and 2010, Brazil's sugarcane production more than doubled – from 326 to 717 million tonnes per harvest – driven by biofuel policies that included rising ethanol mandates and preferential taxation of ethanol relative to gasoline (Cardoso et al., 2019). This expansion was channeled through two distinct ethanol markets. Anhydrous ethanol is blended directly with gasoline under a federally mandated ratio, set at 30% as of August 2025 (CNPE, 2025), and affects fuel consumption symmetrically across all regions. Hydrated ethanol, by contrast, is sold as a standalone fuel and can power any flex-fuel vehicle – which account for more than 80% of Brazil's light vehicle fleet (Anfavea, 2025). Unlike anhydrous ethanol, hydrated ethanol competes directly with gasoline-C at the pump, and that competition is geographically uneven: transport costs erode hydrated ethanol price advantage with distance from the sugarcane belt. In regions far from the producing areas – such as the North – consumers face prices at which hydrated ethanol is no longer competitive with gasoline, effectively suppressing demand.³ Figure A-10 maps this geographic pattern.⁴

[Figure 1]

Three states account for 75% of the sugarcane production in Brazil: São Paulo (52%), Goiás (12%), and Minas Gerais (11%)⁵. São Paulo alone produced 350 million tonnes in 2025 – more than half of Brazil's national output of 673 million tonnes that year (IBGE, 2024)⁶ – across 4.4 million hectares. Were São Paulo a country, its output would nearly

³For a comprehensive overview of ethanol adoption in Brazil, see Fontana et al. (2026).

⁴A further structural shift is underway: corn-based ethanol has grown rapidly and by 2026 accounted for more than 20% of total ethanol production in Brazil (Conab, 2024).

⁵Based on Conab (2024) data for the 2025/2026 harvest.

⁶FAO (2025) reports a slightly higher national total of 717 million tonnes for the same year, reflecting methodological differences in coverage. We use IBGE/PAM figures throughout for within-Brazil comparisons.

match that of India, the world's second-largest producer. Concentration is pronounced within the state as well: production is overwhelmingly located in the northwestern municipalities, while the south contributes negligibly. Figure A-2 shows the share of municipal land allocated to sugarcane cultivation across São Paulo from 2002 to 2010, the pre-treatment period.

[Figure 2]

There is considerable heterogeneity in the intensity of sugarcane cultivation across municipalities. Among the 645 municipalities in São Paulo state, 230 have virtually no sugarcane production (less than 2% of their territory allocated to sugarcane), while 57 municipalities dedicate at least half of their land to sugarcane cultivation (see Figures A-5 and A-6 for the distribution). We exploit this spatial variation in the share of land under sugarcane cultivation as a measure of each municipality's exposure to sugarcane smoke.

2.2 *The Mechanization of Sugarcane Harvesting in São Paulo*

Figures A-7 and A-9 document the shift in São Paulo's sugarcane sector over the past two decades: harvested area expanded sharply through the late 2000s before stabilizing, while the share of mechanically harvested cane rose from roughly one-third in 2007 to near-universality by the mid-2010s.

The state government has emphasized its own role in this transition, pointing to two complementary regulatory interventions. First, the State Law No. 11.241/2002 established a gradual ban on sugarcane straw burning, with targets of 50% of harvested area by 2011, 80% by 2016, and 100% by 2021. Second, the Green Ethanol Protocol,⁷ a voluntary agreement signed in 2007 between the state government and the sugarcane industry, accelerated these deadlines to 2014 and 2017 for mechanizable and other regions, respectively, offering refineries access to preferential financing and environmentally demanding markets in exchange for early compliance.

Davis (2017) challenges the regulatory narrative, arguing that the law explains little of the mechanization observed in São Paulo. His test is geographic: if regulation were the binding constraint, mechanization should be concentrated in São Paulo – where the law applied – yet the data do not clearly support this pattern⁸. Drawing instead on sur-

⁷In Portuguese, *Protocolo Agroambiental Etanol Verde* or *Protocolo Etanol Mais Verde* after 2017.

⁸This test may be complicated by the close economic integration between the north of São Paulo and the south of Minas Gerais – the second-largest producer – which share many of the same companies, agricultural practices, and input suppliers, potentially blurring the regulatory contrast across the state border.

vey evidence, he documents that growers switched to mechanical harvesting primarily in response to rising real wages among low-skilled workers, a pressure that intensified throughout Brazil during the 2000s. This interpretation is further supported by the limited mechanization observed in poorer sugarcane-producing regions of Brazil, where the same wage pressure was absent.

A third driver operates through export market access. Sugar is heavily traded internationally, and buyers – particularly in Europe – increasingly conditioned market access on environmentally responsible production practices. Pre-harvest burning threatened the sustainability credentials of Brazilian sugar at a time when export markets were growing more sensitive to production methods. This demand-side pressure reinforced the incentives created by regulation and labor costs, particularly for mills with significant export exposure (De Moraes and Zilberman, 2014).

The relative weight of these three forces – environmental regulation, rising wages, and export market pressure – remains an open question, and settling it is beyond the scope of this paper. What is clear is that all three contributed to a transition that, by the early 2010s, had fundamentally altered how sugarcane is harvested in São Paulo.

Figure A-3 documents the technical change indirectly: the total area permitted for pre-harvest burning peaked at nearly two million hectares around 2010 and fell to near zero by 2020. Yet a reduction in the area legally permitted for burning need not translate into a reduction in actual fire use if growers do not comply with the policy. Section 5 addresses this concern, showing that municipalities more exposed to the technical change experienced significant reductions in PM_{10} and $PM_{2.5}$ – reductions concentrated exclusively during the harvest season (July–September) and statistically indistinguishable from zero in the off-season. This seasonal pattern is difficult to reconcile with any explanation other than reduced agricultural burning.

3 Data and Empirical Strategy

3.1 Data

Our analysis combines six administrative and environmental datasets for São Paulo state. Summary statistics for all main variables are reported in Table A-2 in the Appendix.

The treatment variable is constructed from the Municipal Agricultural Production

survey (PAM) maintained by IBGE ([IBGE, 2024](#)). We compute the share of each municipality's total area dedicated to sugarcane cultivation, averaged over the pre-treatment period 2002–2010. This pre-period average serves as our primary measure of each municipality's exposure intensity to the technical change. Among the 645 municipalities in São Paulo, roughly half allocate less than 5% of their territory to sugarcane, while the top decile dedicates more than 45%. Figures [A-5](#) and [A-6](#) illustrate how this variable is distributed.

For health outcomes, we draw on three Ministry of Health administrative registries. Hospital admissions come from the Sistema de Informação Hospitalar (SIH/SUS) via Datasus, spanning 2008–2022. We focus on admissions with principal diagnoses in ICD-10 chapters J (respiratory diseases, J00–J99) and I (circulatory diseases, I00–I99), aggregated to municipality-year cells and expressed as rates per capita. Birth records come from the Sistema de Nascidos Vivos (SINASC) via Datasus, covering approximately 9.7 million individual births in São Paulo over 2007–2023. Each record contains birthweight, a congenital anomaly indicator, birth month, maternal education, and municipality of residence. We recode the anomaly indicator from its raw form (1 = yes, 2 = no, 9 = unknown) to a binary variable, treating unknown codes as missing. We also construct a pollution-linked anomaly index (Pol3) equal to one if the birth presents at least one of: neural tube defects (Q00–Q07), orofacial clefts (Q35–Q37), or congenital cardiac malformations (Q20–Q26). Mortality records come from the Sistema de Informações sobre Mortalidade (SIM/Datasus), covering 2002–2023. We use respiratory and circulatory cause-specific mortality rates per capita for adults, and respiratory mortality per 1,000 live births for children under five.

To document the smoke channel, we combine two independent data sources. Ground-level air quality data come from daily PM₁₀ and PM_{2.5} readings at Cetesb (Companhia Ambiental do Estado de São Paulo) monitoring stations, available over 2003–2024, which we aggregate to municipality-season-year cells. Fire incidence is measured by cross-referencing the MODIS MCD64A1 monthly burned-area product (500-meter resolution, Collection 6.1) with the MapBiomas Land Use and Land Cover sugarcane layer (Collection 9, 30-meter resolution), both processed via Google Earth Engine. This cross-referencing allows us to isolate burned area within sugarcane pixels and to compare the harvest season (July–September) against the off-season (January–March) as a placebo.

The wind direction triple-difference relies on a municipality-level upwind sugarcane exposure measure constructed from Nasa Power/Merra-2 reanalysis wind data for July–

September, 2002–2022 (Chagas et al., 2016). For municipality i in year t :

$$WindExp_{it} = \sum_j \frac{\max(0, \cos(\hat{w}_{it} - \angle_{i \rightarrow j}))}{dist_{ij}} \cdot SC_j, \quad dist_{ij} \leq 100 \text{ km}$$

where $\hat{w}_{it}$ is the prevailing wind direction at i , $\angle_{i \rightarrow j}$ is the bearing from i toward source municipality j , and SC_j is j 's sugarcane share. Because wind direction varies from year to year, $WindExp_{it}$ has within-municipality time variation and is not collinear with the municipality fixed effects. We also collect population estimates from IBGE and GDP per capita from IBGE municipal accounts, used as regression weights and controls, respectively.

3.2 Empirical Strategy

Our primary design exploits the timing of the burning reduction and the spatial variation in sugarcane land use:

$$\ln(Y_{it}) = \alpha \cdot TC_{it} + \theta_i + \gamma_t + \epsilon_{it} \quad (1)$$

where Y_{it} is the health outcome of municipality i in year t ; θ_i and γ_t are municipality and year fixed effects; and $TC_{it} = Land_i \times D_{t>2010}$ is the exposure to the technical change—the product of the pre-treatment sugarcane land share and a dummy equal to one after 2010. α is our parameter of interest. Regressions are weighted by population (or by births for birth outcomes), and standard errors are clustered at the municipality level. For individual-level birth outcomes, we estimate the analogous linear probability model (LPM), replacing municipality-year observations with individual births and adding fixed effects for maternal education, sex, and marital status.

We augment the baseline specification with three strategies that sharpen identification. The first is a harvest season triple-difference, which exploits the fact that sugarcane burning – and thus PM exposure – is concentrated in July-September:

$$\ln Y_{itm} = \alpha \cdot TC_{it} + \beta \cdot TC_{it} \times Harvest_m + \theta_{i \times m} + \gamma_t + \epsilon_{itm} \quad (2)$$

where $Harvest_m = \mathbf{1}(m \in \{7, 8, 9\})$ and $\theta_{i \times m}$ are municipality $\times$ month fixed effects that absorb local seasonal patterns. The coefficient β captures the additional effect of exposure during burning months. The second strategy adds the time-varying $WindExp_{it}$ variable to

equation (1), testing whether municipalities downwind of sugarcane areas benefit more from the transition. The third replaces the binary $D_{t>2010}$ indicator with $fase_t \in [0, 1]$ —the fraction of pre-2011 baseline sugarcane burning emissions already eliminated in year t , constructed from SEEG data – and interacts it with the municipal sugarcane land share. Because $fase_t$ is common to all municipalities in a given year, it provides an exogenous state-level shifter that scales with the actual pace of the emissions reduction.

4 Results

4.1 Hospitalizations

Table 1 presents results for respiratory and circulatory hospitalizations. Technical change exposure reduces hospitalizations across all specifications and both disease categories, with respiratory outcomes responding more strongly than circulatory ones – consistent with the primarily airborne nature of the smoke channel. The direction of the effect is intuitive: municipalities more intensive in sugarcane production were more exposed to pre-harvest burning and, consequently, stand to benefit more from its elimination.

Specifically, for respiratory diseases (Panel A), a one-standard-deviation increase in exposure ($SD = 0.19$) is associated with a 7.2% reduction in per capita hospitalizations (0.380×0.19). The estimates are robust to controlling for per capita GDP and to excluding the COVID-19 period. Pre-COVID coefficients are slightly smaller in magnitude (-0.287 vs. -0.380) but statistically indistinguishable from the full-sample estimates. Controlling for GDP leaves the point estimates virtually unchanged – suggesting that the health improvements are not driven by differential economic trends across municipalities.

For circulatory diseases (Panel B), a one-standard-deviation increase in exposure reduces hospitalizations by 4.5%. Across all specifications, the circulatory point estimates are slight smaller than their respiratory counterparts. The circulatory estimates are fully stable across both the full sample and the pre-COVID period, and controlling for GDP leaves the point estimates essentially unchanged, as with respiratory diseases.

4.2 Birth Outcomes

Table 2 presents results from the individual-level analysis, estimated on approximately 9.6 million SINASC birth records from São Paulo (2007–2023). The specification extends the

Table 1: Hospitalizations Primarily Caused by Respiratory and Circulatory Diseases

	$y = \ln(\text{hospitalizations per capita})$			
	(1)	(2)	(3)	(4)
	Full	Full+GDP	Pre-Covid	Pre-Covid+GDP
<i>Panel A — Respiratory Diseases (ICD J00–J99)</i>				
Tech. Change (α)	−0.380*** (0.114)	−0.369*** (0.112)	−0.287** (0.111)	−0.286** (0.111)
ln(GDP)		−0.146* (0.083)		−0.036 (0.071)
Observations	4,739	4,739	4,081	4,081
<i>Panel B — Circulatory Diseases (ICD I00–I99)</i>				
Tech. Change (α)	−0.238*** (0.078)	−0.224*** (0.075)	−0.234*** (0.078)	−0.222*** (0.075)
ln(GDP)		−0.078 (0.063)		−0.057 (0.063)
Observations	4,689	4,689	4,039	4,039
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Weights	Pop.	Pop.	Pop.	Pop.
Period	2008–2021	2008–2021	2008–2019	2008–2019

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the logarithm of hospitalizations per capita primarily caused by respiratory diseases (Panel A) or circulatory diseases (Panel B). Pre-Covid columns exclude observations after 2019.

baseline design with additional fixed effects for maternal education, sex, and marital status. Technical change exposure has no detectable effect on birthweight outcomes – coefficients for LBW and VLBW are very small and statistically indistinguishable from zero ($p = 0.535$ and $p = 0.902$, respectively). It is consistent with the interpretation that exposure to the technical change lies below the threshold for general fetal growth restriction.

The technical change does, however, matter for developmental processes during critical windows of organogenesis. Greater exposure significantly reduces both overall congenital anomalies and the pollution-linked anomaly index. For the overall anomaly indicator, a one-standard-deviation increase in exposure ($SD = 0.19$) is associated with an 11% relative

Table 2: Birth Outcomes — Individual-Level Linear Probability Model

	Congenital Anomalies		Birthweight	
	(1) Any Anomaly (Q00–Q99)	(2) Pol3 Index (NTD + Orofac. + Card.)	(3) LBW (<2,500g)	(4) VLBW (<1,500g)
Tech. Change (α)	−0.00666* (0.00390)	−0.00427** (0.00182)	0.00124 (0.00200)	0.00010 (0.00084)
Observations	9,551,237	9,599,799	9,599,799	9,599,799
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Education FE	✓	✓	✓	✓
Sex FE	✓	✓	✓	✓
Marital status FE	✓	✓	✓	✓
Baseline mean	0.01145	0.00359	0.09264	0.01472

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Individual SINASC birth records for São Paulo, 2007–2023. Column (1): any congenital anomaly (ICD-10 Q00–Q99). Column (2): Pol3 index, equal to one if the birth presents at least one of: neural tube defects (Q00–Q07), orofacial clefts (Q35–Q37), or congenital cardiac malformations (Q20–Q26). Columns (3)–(4): birthweight below 2,500 g and 1,500 g. $TC_{m,t} = ShareCane_m \times \mathbf{1}(t > 2010) \times \mathbf{1}(ShareCane_m > 0.05)$.

reduction from a baseline of 11.5 per 1,000 births ($p < 0.10$). The effect is more precisely estimated for the pollution-linked index: a one-standard-deviation increase reduces its incidence by 22.6% relative to a baseline of 3.59 per 1,000 births.

Table 3 disaggregates the pollution-linked index by diagnostic group. Cardiac malformations drive the largest effect in both absolute and relative terms: the coefficient of -0.003846^{**} implies a 38.3% relative reduction at one standard deviation of exposure, from a baseline of 1.91 per 1,000 births. This finding is consistent with established biological pathways – fine particulate matter and combustion gases generate reactive oxygen species that cross the placental barrier and disrupt conotruncal septation. Neural tube defects contribute a smaller but significant reduction of 9.4% at one standard deviation. Orofacial clefts show a coefficient of effectively zero ($p = 0.999$), consistent with their primary teratogenic pathway running through organophosphate pesticide exposure rather than combustion-derived PM – and sugarcane cultivation is not pesticide-intensive in the relevant sense.

Finally, we find no socioeconomic gradient in the anomaly reduction. Repeating the

Table 3: Congenital Anomalies by Diagnostic Group

$y = \text{Pr}(\text{anomaly type})$	(1) Neural Tube (Q00–Q07)	(2) Orofacial (Q35–Q37)	(3) Cardiac (Q20–Q26)	(4) Pol3 Index
Tech. Change (α)	−0.000539** (0.000221)	0.000000 (0.000156)	−0.003846** (0.001623)	−0.004265** (0.001823)
Observations	9,599,799			
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Education FE	✓	✓	✓	✓
Sex FE	✓	✓	✓	✓
Marital status FE	✓	✓	✓	✓
Baseline (per 1,000)	1.09	0.73	1.91	3.59

Standard errors clustered by municipality in parentheses. ** $p < 0.05$. Each column is a separate LPM regression with municipality, year, education, sex, and marital status fixed effects.

results for sub-samples by maternal education group yields coefficients that are similar in magnitude and have fully overlapping standard errors across all groups. This suggests that the anomaly reduction reflects broad ambient PM exposure that benefits fetuses across the income distribution, rather than differential access to protective behaviors or healthcare.

4.3 Mortality

Table 4 presents results for adult respiratory and circulatory mortality. Across all four specifications, the coefficients on Exp are negative – signed in the expected direction – but negligible in magnitude and statistically indistinguishable from zero. At one standard deviation of exposure ($SD = 0.19$), the point estimates imply reductions of only 0.5% in respiratory mortality (0.028×0.19) and 0.8% in circulatory mortality (0.043×0.19) – effects too small to be distinguished from noise.

The null result is informative. Mortality is the least sensitive margin in the pollution–health dose–response relationship, and the particulate matter reductions achieved during the harvest season may fall below the threshold required to produce detectable effects on adult mortality. The more sensitive margin – infant respiratory mortality – tells a different story.

Table 4: Mortality Primarily Caused by Respiratory and Circulatory Diseases

	Respiratory (J00–J99)		Circulatory (I00–I99)	
$y = \ln(\text{mortality rate})$	(1)	(2)	(3)	(4)
Exp	−0.028 (0.093)	−0.029 (0.092)	−0.043 (0.071)	−0.049 (0.067)
$\ln(\text{GDP})$		0.039 (0.039)		0.091 (0.055)
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Weights	Pop.	Pop.	Pop.	Pop.
Period	2002–2023			

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the logarithm of the mortality rate for respiratory (columns 1–2) or circulatory (columns 3–4) causes.

Table 5 restricts the mortality analysis to infant and child age groups. Panel A shows that infant respiratory mortality – deaths from respiratory causes (ICD J00–J99) among children under five, expressed per 1,000 live births – responds to the technical change. A one-standard-deviation increase in exposure ($SD = 0.19$) is associated with an approximately 11% reduction in the infant respiratory mortality rate (0.593×0.19), significant at the 10% level across the full and pre-COVID samples, and at the 5% level when controlling for per capita GDP. All-cause infant mortality (Panel B) shows no effect across all specifications, consistent with the reduction operating through the respiratory channel rather than reflecting a broad improvement in general infant survival. Respiratory mortality among children aged 5–11 years (Panel C) is also indistinguishable from zero, likely reflecting insufficient statistical power given the rarity of respiratory deaths in this age group at the municipality level.

The contrast between null adult mortality and significant infant respiratory mortality is informative about the dose-response gradient. Infants are physiologically more susceptible to particulate matter: narrower airways, higher respiratory rates, and lower detoxification capacity make them sensitive at pollution thresholds below those required to affect adult mortality (Saravia et al., 2013). The reduction in PM_{10} and $PM_{2.5}$ during the harvest season is sufficient to shift the infant respiratory mortality distribution, even while falling short of the threshold for adult effects.

Table 5: Infant and Child Mortality by Cause and Age Group

	$y = \ln(\text{mortality rate per 1,000 births} + 0.001)$			
	(1)	(2)	(3)	(4)
	2001–2023	2001–2019	2001–2023	2001–2019
			+GDP	+GDP
<i>Panel A — Infant respiratory mortality (J00–J99, <5 years)</i>				
Tech. Change (α)	−0.593*	−0.629*	−0.765**	−0.622
	(0.344)	(0.367)	(0.378)	(0.384)
<i>Panel B — Infant all-cause mortality (<5 years)</i>				
Tech. Change (α)	−0.053	−0.064	−0.049	−0.053
	(0.114)	(0.121)	(0.117)	(0.121)
<i>Panel C — Children 5–11 years, respiratory mortality</i>				
Tech. Change (α)	−0.075	−0.222	−0.376	−0.350
	(0.639)	(0.542)	(0.705)	(0.642)
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Weights	Births	Births	Births	Births

Standard errors clustered by municipality in parentheses. * $p < 0.1$, ** $p < 0.05$. Outcome: $\ln(\text{cause-specific mortality rate per 1,000 live births} + 0.001)$. Regressions weighted by live births per municipality-year. Sample: São Paulo municipalities, 2001–2023 (full) and 2001–2019 (pre-Covid).

5 The Pollution Channel

One natural concern is whether the health improvements documented above flow through the smoke-reduction channel rather than any confounding factor. We present three pieces of evidence that support the smoke interpretation. First, satellite data confirm that fire use on sugarcane fields fell sharply in municipalities more exposed to the technical change – and only during the harvest season, when burning would have occurred. Second, ground-level particulate matter concentrations fell during those same months, with no detectable effect in the off-season. Third, hospitalization reductions are disproportionately concentrated in harvest months, consistent with an acute smoke-exposure mechanism rather than a permanent shift in local health conditions.

To confirm that the technical change reduced burning specifically within sugarcane fields – rather than displacing fire to other land uses or seasons – we cross-reference satellite burned-area data with sugarcane land cover, computing burned area separately for the harvest season (July–September) and the off-season (January–March).⁹

Virtually all sugarcane burning occurs during the harvest season; off-season burned area remains essentially flat near zero throughout the entire period, providing a natural placebo that recurs in the particulate-matter results below. Panel A of Table 6 quantifies the harvest-season decline within municipalities, controlling for common temporal shocks through municipality and year fixed effects. The outcome is $\ln(\text{area burned in sugarcane pixels} + 0.001)$ during the harvest season. Municipalities more exposed to the technical change experienced a reduction of 4.96 log points per unit of exposure ($p < 0.01$, $SE = 0.45$). At the pre-mechanization mean sugarcane land share ($\overline{\text{share_cana}} \approx 0.22$), this corresponds to approximately a 66% reduction in harvest-season burned area relative to the pre-mechanization baseline. This confirms that higher exposure translated into less burning exactly where and when the transition required.

Panel B of Table 6 turns from fire to air quality, showing that the burning reduction transmitted directly into lower PM concentrations. Using daily PM_{10} and $PM_{2.5}$ readings from Cetesb monitoring stations, aggregated to municipality-season-year cells, we estimate equation (1) separately for harvest and off-season months.

At one standard deviation of municipal sugarcane land share ($SD = 0.19$), the technical change reduced PM_{10} by 1.8% and $PM_{2.5}$ by 1.7% during the harvest season (0.094×0.19 and 0.091×0.19 , respectively). The off-season coefficients are statistically indistinguishable from zero ($p > 0.4$ for PM_{10} ; $p > 0.8$ for $PM_{2.5}$): concentrations fall by only 0.5% and 0.1%, respectively, in January–March. This seasonal contrast is another evidence for the smoke mechanism. Panels A and B together establish the first two links: the technical change reduced actual fire use on sugarcane fields, and this translated into lower particulate matter concentrations during the months when burning occurs.

The third link connects reduced PM to reduced hospitalizations. Table 7 presents estimates of equation (2), which exploits monthly variation in hospitalizations to test whether health improvements concentrate in the burning season.

For respiratory hospitalizations, the year-round effect is -0.135^{***} , and the additional

⁹Sugarcane pixels are identified using MapBiomass Land Use and Land Cover Collection 9 (class 20, 30-meter resolution). Burned-area detection relies on the MODIS MCD64A1 Collection 6.1 monthly burned-area product (500-meter resolution). Both layers are processed via Google Earth Engine.

Table 6: The Pollution Channel: Sugarcane Burning and Particulate Matter

	Harvest (Jul–Sep)	Off-season (Jan–Mar)
<i>Panel A: Sugarcane Burning (MODIS $\times$ MapBiomass)</i>		
ln(Burned area in sugarcane, ha)	−4.964*** (0.450)	—
Observations	15,216	
<i>Panel B: Particulate Matter (CETESB)</i>		
ln(PM10)	−0.094*** (0.010)	−0.005 (0.006)
ln(PM25)	−0.091*** (0.010)	−0.001 (0.006)
Observations	14,124	14,124
Municipality FE	✓	✓
Year FE	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors clustered at the municipality level. Panel A: outcome is $\ln(\text{area burned in sugarcane pixels} + 0.001)$, harvest season (July–September). Off-season column omitted: the $\log(\cdot + 0.001)$ specification is unreliable when zeros dominate, as they do in January–March. Source: MODIS MCD64A1 burned-area product crossed with MapBiomass LULC sugarcane layer (Collection 9), processed in Google Earth Engine. Panel B: daily CETESB station data aggregated to municipality-season-year, 2003–2024.

harvest-season increment is a highly significant -0.069^{***} : the respiratory reduction concentrates in July–September, exactly when burning would have occurred. For circulatory hospitalizations, the off-season baseline effect is -0.083^{**} , but the harvest-season increment is small and insignificant (-0.014), consistent with the circulatory channel being more diffuse and less directly linked to acute airway exposure. Across all three data sources – satellite fire detection, ground-level PM monitoring, and monthly hospitalization records – the evidence points to the same conclusion: the health improvements documented in Section 4 flow through the smoke-reduction channel.

Table 7: Harvest Season Triple-Difference for Hospitalizations

$y = \ln(\text{hospitalizations/pop})$	Respiratory	Circulatory
Exp (off-season baseline)	−0.135*** (0.033)	−0.083** (0.028)
$Exp \times \text{Harvest (Jul-Sep)}$	−0.069*** (0.011)	−0.014 (0.013)
Municipality $\times$ Month FE	✓	✓
Year FE	✓	✓
Weights	Pop.	Pop.

** $p < 0.05$, *** $p < 0.01$. Municipality $\times$ month fixed effects absorb local seasonal patterns in hospitalization rates. Exp captures the year-round effect; $Exp \times \text{Harvest}$ captures the additional effect during burning months (July–September).

6 Identification and Robustness

6.1 Parallel Trends and Event Study

We test the parallel trends assumption by estimating a dynamic version of equation (1):

$$\ln Y_{it} = \sum_{\tau \neq -1} \beta_{\tau} (D_{\tau} \times SC_i) + \theta_i + \gamma_t + \varepsilon_{it}$$

where D_{τ} is an indicator for year τ relative to 2010 (the reference year). Table 8 shows selected coefficients.

Table 8: Event Study Coefficients — Selected Years

Year (τ relative to 2010)	Respiratory	Circulatory
$\tau = -3$ (2008)	+0.028 (0.091)	+0.148 (0.088)
$\tau = -2$ (2009)	+0.110 (0.047)	+0.092 (0.056)
$\tau = -1$ (2010)	0 (reference)	
$\tau = 0$ (2011)	−0.116	−0.136
$\tau = 5$ (2016)	−0.335	−0.255
$\tau = 10$ (2021)	−0.778	−0.271

Pre-treatment coefficients are near zero and statistically insignificant, supporting the parallel trends assumption. Full event study figures are shown in Figures A-11 and A-12 in the Appendix.

Pre-treatment coefficients are small and statistically indistinguishable from zero, supporting the validity of the parallel trends assumption. Post-treatment coefficients grow steadily in magnitude over time, consistent with the gradual and accelerating elimination of burning under the GEP.

6.2 Wind Direction Triple-Difference

Table 9 presents results augmenting equation (1) with $WindExp_{it}$, which measures the degree to which a municipality is downwind of sugarcane areas during the harvest season (see Section 3 for the construction). If the technical change improves health through reduced smoke, municipalities downwind of sugarcane areas should benefit disproportionately.

Table 9: Wind Direction Triple-Difference

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Exp	−0.348*** (0.100)	−0.265*** (0.097)	−0.238*** (0.078)	−0.222*** (0.078)
$WindExp$	−0.035* (0.019)	−0.027 (0.021)	−0.033 (0.021)	−0.011 (0.022)
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

* $p < 0.1$, *** $p < 0.01$. Clustered SEs (municipality). $WindExp_{it}$ is the directional exposure to upwind sugarcane burning, constructed from NASA POWER/MERRA-2 wind data (July–September). Figure A-14 maps $WindExp$ across municipalities.

The main effect of Exp remains negative and significant across all columns, reassuringly close to the baseline estimates. The $WindExp$ coefficient is negative – municipalities more downwind of sugarcane areas benefit more from the technical change – and is significant at the 10% level for respiratory hospitalizations in the full sample. For circulatory hospitalizations, the sign is consistent but imprecise. This pattern provides additional evidence that the health improvements operate through the airborne smoke channel.

6.3 Placebo: All ICD-10 Diagnostic Groups

We treat injuries and trauma (group S), transport accidents (group V), and falls and drowning (group W) as placebo outcomes because they have no plausible biological link to sugarcane smoke. One might argue, however, that this choice is convenient. To address this concern directly, we estimate the same baseline specification for every ICD-10 diagnostic group (A through Z) and collect all coefficients in a single figure. For each chapter c , we estimate:

$$\ln(\text{hospitalizations per capita}_{it}^c + 0.001) = \alpha_i + \alpha_t + \beta^c \cdot \text{Exp}_{it} + \varepsilon_{it}$$

Figure A-16 shows the results. Two patterns stand out. First, the largest negative effects concentrate in the diagnostic groups with a direct mechanistic link to particulate matter: respiratory (J) and circulatory (I) diseases. Second, the three groups we selected as placebos – S, V, and W – are statistically indistinguishable from zero, validating the original placebo choice.

The majority of the remaining groups are also indistinguishable from zero. Four show significant negative coefficients – infectious and parasitic (A), blood and immune (D), nervous system (G), and digestive (K) – a pattern consistent with the broader systemic effects of $\text{PM}_{2.5}$ documented in the literature, as fine particles trigger inflammatory cascades that extend beyond the respiratory tract (?). Establishing causality for each of these groups is beyond the scope of this paper. Crucially, the overall pattern does not support a confounding interpretation: any confounder driving the respiratory and circulatory results would need to leave trauma, mental health, neoplasms, pregnancy, and most other diagnostic groups entirely unaffected – a highly implausible requirement.

6.4 Continuous Treatment Intensity

The binary specification assumes full treatment from 2011 onward; in reality, the GEP phased in gradually. We replace $\mathbf{1}(t > 2010)$ with $f_{ase_t} \in [0, 1]$, defined as the fraction of pre-2011 baseline sugarcane burning emissions already eliminated by year t (from SEEG data):

$$f_{ase_t} = \frac{\text{Baseline}_{SP} - \text{Emiss}_{SP,t}}{\text{Baseline}_{SP}}$$

This variable is common to all municipalities in a given year, providing an exogenous state-level shifter that interacts with the municipal-level sugarcane land share. Table 10

presents results alongside the binary estimates for comparison.

Table 10: Continuous Treatment Intensity

	Respiratory	Circulatory
Binary TC (baseline)	-0.344^{***} (0.087)	-0.239^{***} (0.078)
Continuous TC ($\times f_{ase_t}$)	-0.618^{***} (0.168)	-0.393^{**} (0.152)
Municipality FE	✓	✓
Year FE	✓	✓
Weights	Pop.	Pop.

** $p < 0.05$, *** $p < 0.01$. The binary and continuous specifications are estimated separately. Implied effect at mean $f_{ase_t} \approx 0.56$: $-0.618 \times 0.56 \approx -0.346$, nearly identical to the binary estimate of -0.344 . Figure A-15 plots f_{ase_t} over time.

The continuous treatment estimates are larger in magnitude— -0.618 for respiratory and -0.393 for circulatory hospitalizations—but at the mean value of $f_{ase_t} \approx 0.56$ imply effects nearly identical to the binary estimates (-0.346 vs. -0.344). This confirms that the binary specification captures the average effect correctly, and that the health improvements build up proportionally with the actual pace of the emissions reduction.

6.5 Excluding the City of São Paulo

São Paulo city, the state capital, has a population of approximately 12 million but essentially zero sugarcane land share, placing it permanently in the control group. Its size means it can dominate population-weighted regressions. Table 11 shows that results are robust to its exclusion.

Coefficients remain negative and statistically significant after removing São Paulo city. The modest reduction in magnitude is expected: the large control-group city contributes precision to the baseline estimate, and its removal leaves slightly less power.

Table 11: Robustness: Excluding the City of São Paulo

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Baseline	−0.380***	−0.287**	−0.238***	−0.234***
Excl. SP city	−0.327**	−0.236*	−0.228**	−0.189**

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row reports the coefficient on TC from a separate regression; the excluded municipality is IBGE code 3550308.

7 Discussion

7.1 Cost Savings from Mechanization

Lower hospitalization rates and reduced congenital anomaly incidence represent direct cost savings to the Brazilian public health system. To quantify these savings conservatively, we multiply the estimated reduction in each outcome by the average 2019 SUS reimbursement cost for that hospitalization category in São Paulo, using a single reference year to avoid inflation-adjustment assumptions.¹⁰

For respiratory and circulatory hospitalizations, a one-standard-deviation increase in exposure reduces hospitalizations by 7.2% and 4.5%, respectively. Aggregating the implied reduction across all exposed municipalities over 2011–2021 yields approximately 118,000 fewer hospital admissions—of which 71,000 are respiratory and 47,000 are circulatory. At average 2019 SUS reimbursement costs of R\$ 1,134 and R\$ 2,739 per admission for respiratory and circulatory diseases, this translates into R\$ 209 million (USD 53 million) in avoided hospitalization expenditures over the decade.

The mechanization also reduced the incidence of congenital cardiac malformations (ICD-10 Q20–Q26), the anomaly type with the strongest point estimate in our analysis. At the estimated coefficient of -0.003846 per birth (Table 3), we estimate approximately 1,640 fewer births with cardiac malformations across exposed municipalities over 2011–2023. Each such birth generates at minimum one surgical hospitalization: at an average 2019 SUS reimbursement of R\$ 13,622 for cardiac malformation admissions, this lower bound corresponds to R\$ 22 million (USD 6 million) in avoided costs.

¹⁰All figures are expressed in 2019 Brazilian reals. In 2019, the average BRL/USD exchange rate was approximately R\$ 3.94 per dollar (PTAX/BCB annual average).

Together, the avoided hospitalization and cardiac anomaly costs amount to R\$ 232 million (USD 59 million) in 2019 reals. This figure is a very conservative lower bound. It excludes outpatient visits, pharmaceuticals, productivity losses, and the costs associated with neural tube defects and orofacial clefts – the other two components of the Pol3 index. It also excludes the BPC/LOAS social protection transfer (approximately one minimum wage per month, lifetime) that accrues to children with permanent disability resulting from severe cardiac malformations. The value of infant lives saved through reduced respiratory mortality – discussed below – represents an additional dimension of welfare improvement not captured here.

8 Final Remarks

The pattern of effects across outcomes is itself informative about the biological mechanisms at work. Adult mortality is the least sensitive margin in the pollution–health dose–response relationship: the PM reductions achieved during the harvest season fall below the threshold required to produce detectable effects on adult mortality. Infant respiratory mortality, by contrast, responds at pollution levels that adult physiology tolerates without measurable harm – consistent with the lower detoxification capacity and higher respiratory rates of children under five. The congenital anomaly results reveal a third, distinct margin: developmental damage during organogenesis that leaves no trace in birthweight yet reduces cardiac malformations and neural tube defects. This hierarchy – null adult mortality, significant infant respiratory mortality, significant hospitalizations, significant anomalies – traces a dose-response gradient in which the same environmental shock propagates differently across physiological margins defined by distinct vulnerability thresholds.

Two of the outcomes we document have not previously been examined in the sugarcane burning literature. Cardiac malformations and neural tube defects have established biological pathways to combustion-derived particulate matter ([Tanwar et al., 2017](#); [Chen et al., 2024](#)), yet no prior study has linked sugarcane burning to these outcomes with causal identification. Our estimates fill this gap by exploiting a large-scale, plausibly exogenous transition in harvesting technology – a setting that approximates the conditions required for credible causal inference in ways that purely observational approaches cannot.

Our estimates are best understood as lower bounds on the true health cost of pre-harvest burning. Two mechanisms operate in this direction. First, *residential sorting*: individuals with greater sensitivity to air pollution may disproportionately avoid or exit municipalities

with intensive sugarcane production, leaving behind a population selected toward lower sensitivity. Second, *avoidance behavior*: residents of cane-intensive municipalities may partially offset ambient exposure through protective strategies – air conditioning, window closing, reduced outdoor activity during the harvest season. Both mechanisms compress the measured treatment effect relative to the counterfactual of full, unmitigated exposure. This compression carries a policy-relevant implication. If the health costs of burning are systematically underestimated because the most sensitive individuals have already sorted away or adapted, then the social returns to clean production exceed what private incentives reflect – and the case for policy intervention is correspondingly stronger.

Even under this conservative interpretation, the magnitudes are substantial. Aggregating across exposed municipalities over the treatment period, we estimate approximately 118,000 fewer hospital admissions and 1,640 fewer births with congenital cardiac malformations, corresponding to R\$ 232 million (USD 59 million, in 2019 reais) in avoided costs at 2019 SUS reimbursement rates. This figure excludes outpatient visits, pharmaceuticals, productivity losses, the lifetime social protection transfers associated with severe birth defects, and the value of infant lives saved through reduced respiratory mortality – all of which would substantially raise the total.

These findings carry direct implications for regions and countries where agricultural burning persists. Within Brazil, the transition remains incomplete: northeastern states, including Pernambuco, still rely predominantly on manual harvesting, with over 90% of sugarcane cut by hand ([Conab, 2024](#)). The GEP offers a replicable institutional model – a voluntary agreement that converted sustainability credentials into preferential access to export markets with high environmental standards, aligning mill incentives with early adoption of cleaner technology. Extending this model to the Northeast, where labor market conditions and export exposure differ, represents a direct and quantifiable policy opportunity. Globally, pre-harvest burning persists in India, Thailand, and South Florida, where burning continues despite full mechanization of the cutting process. For policymakers in these settings, our estimates provide evidence on the health externalities that domestic regulation – or international supply chain sustainability standards – should price in.

One dimension of the transition that our analysis does not capture is the labor market adjustment. Mechanization displaced a substantial share of the manual cane-cutting workforce – a segment concentrated among low-skilled workers. Whether the health gains documented here are sufficient to offset the transition costs borne by displaced workers is a distributional question our analysis cannot resolve. Quantifying the full welfare calculus of agricultural mechanization – incorporating health improvements, income losses,

and distributional consequences – is a natural extension that the richness of Brazilian administrative data makes tractable. Whether newer programs such as RenovaBio – which issues decarbonization certificates to producers who reduce emissions, including through the elimination of pre-harvest burning – can replicate the institutional dynamic of the GEP in regions where structural barriers remain binding is an open question that our framework is well-suited to address as the program matures.

More broadly, this paper demonstrates that large-scale agricultural transitions can generate measurable health dividends – and that these dividends are largest precisely where physiological vulnerability is highest. Practices that reduce short-run production costs but impose health externalities on surrounding populations create a systematic gap between private and social returns. Closing that gap requires both rigorous measurement of the costs imposed and institutional arrangements – whether regulatory mandates, voluntary certification schemes, or market-access conditions – capable of translating those costs back into producer incentives. São Paulo’s transition from fire to mechanical harvesting offers one model of how that alignment can be achieved.

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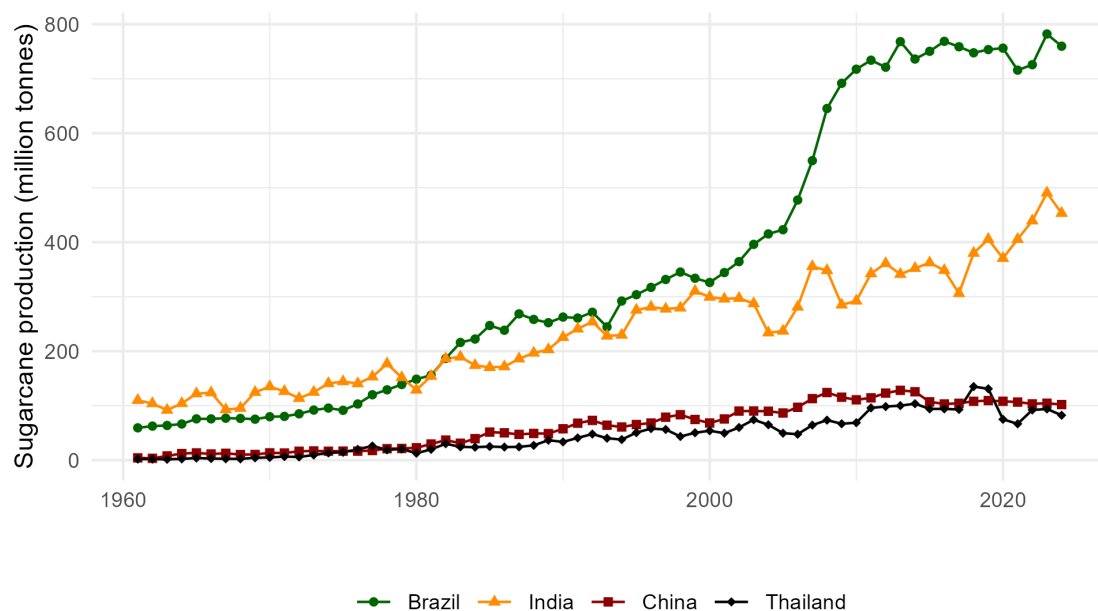
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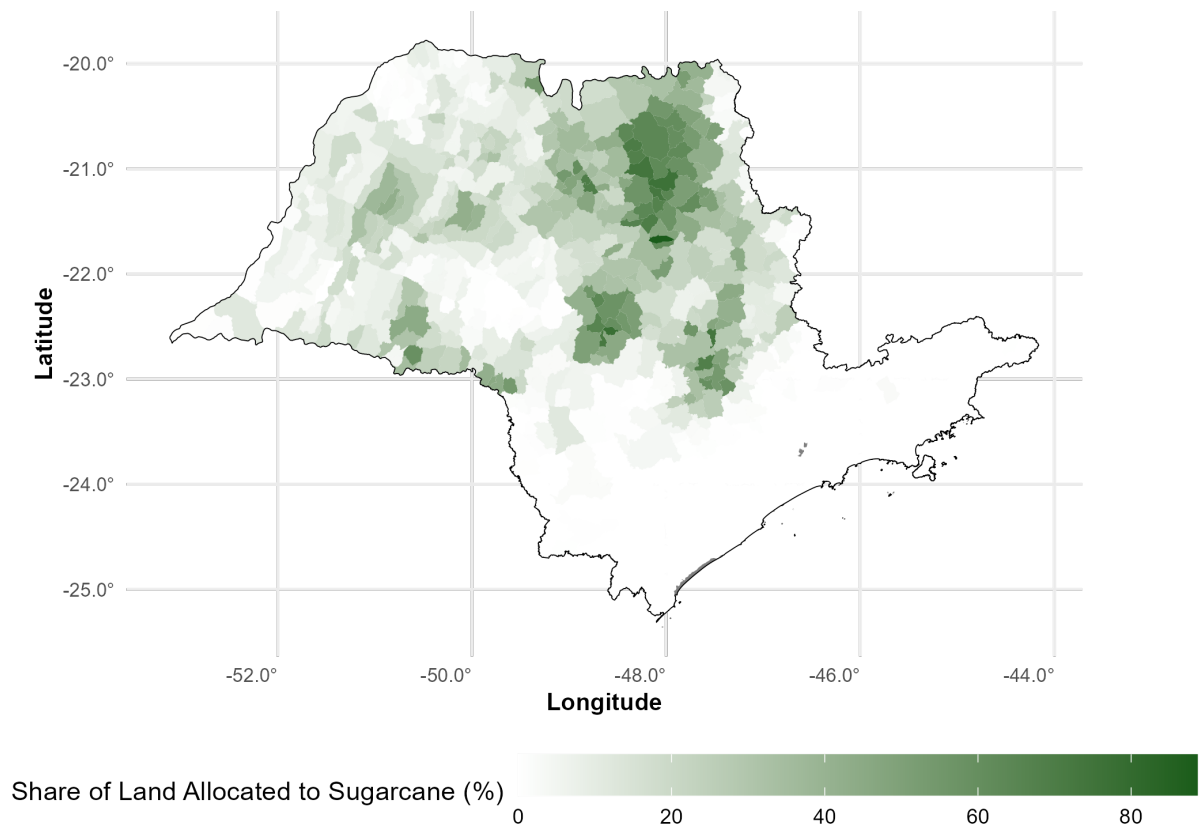
Appendix

Figure A-1: Sugarcane Production Trends for the Top 4 Producing Countries



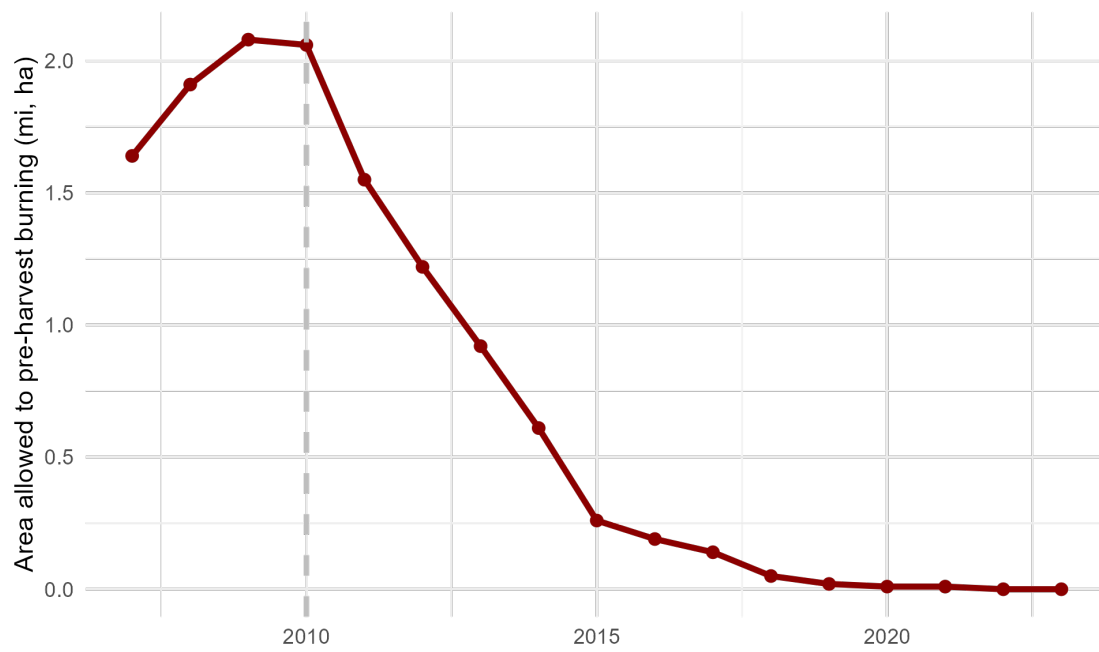
Author's elaboration using data from [FAO \(2025\)](#), with major processing by Our World in Data. The figure displays annual sugarcane production in million tonnes for the four largest producing countries.

Figure A-2: Spatial Distribution of Land Allocated to Sugarcane Cultivation in São Paulo (2002–2010)



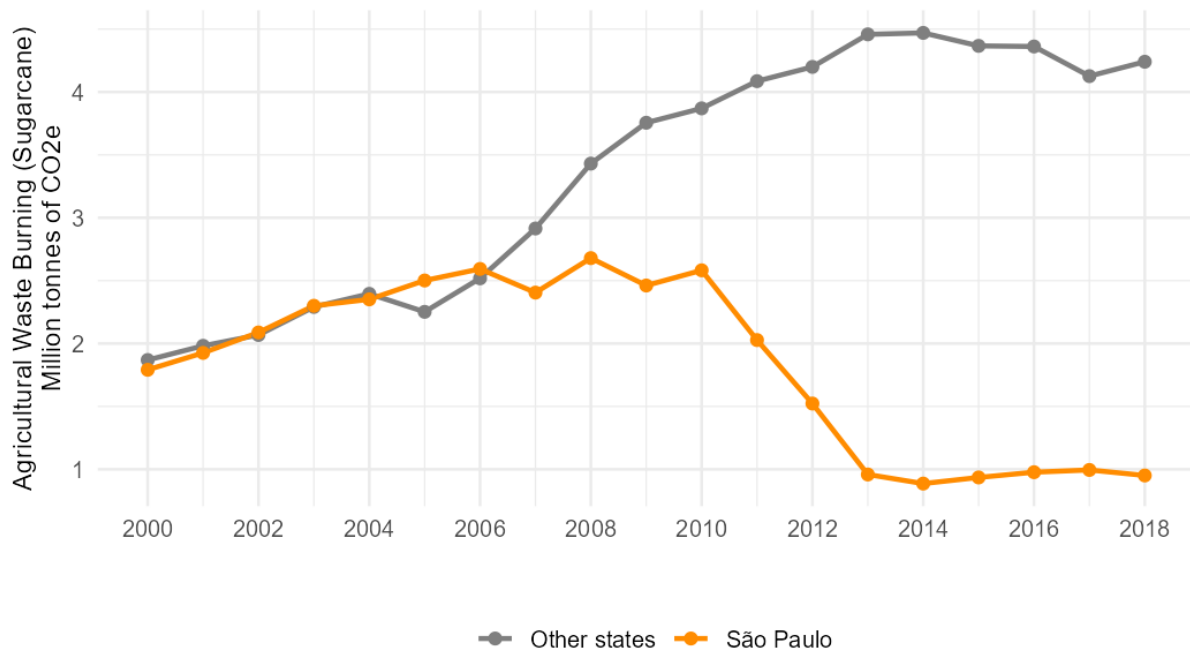
Data from IBGE-Sidra, Pesquisa Agrícola Municipal ([IBGE, 2024](#)). The figure shows the share of land allocated to sugarcane cultivation in each municipality between 2002 and 2010.

Figure A-3: Land Permitted for Pre-Harvest Burning in São Paulo



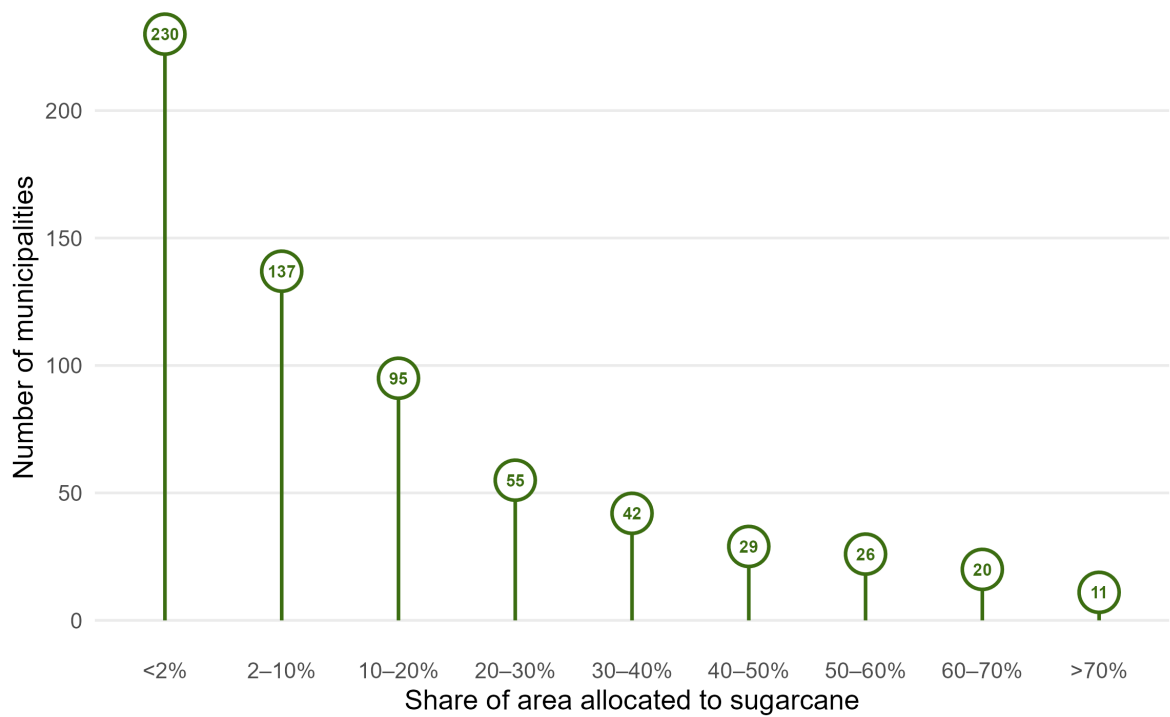
Data from [SAA-SP \(2023\)](#). The graph illustrates total area permitted for pre-harvest sugarcane burning. The gray dashed line marks when the significant reduction began.

Figure A-4: Sugarcane Burning Emissions in São Paulo State (SEEG)



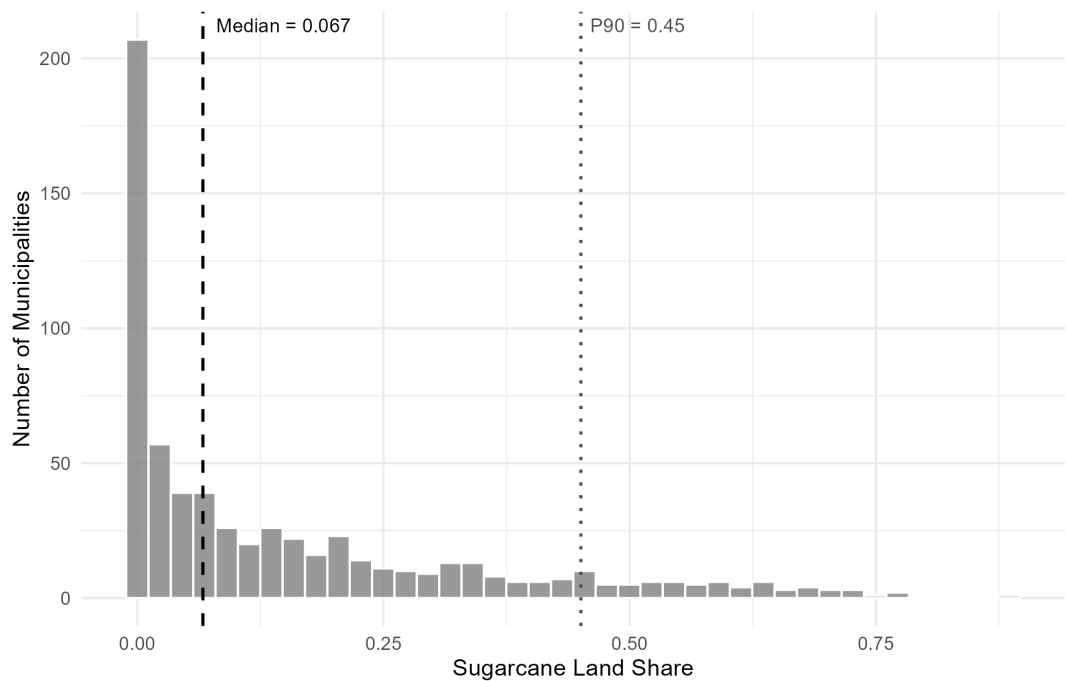
Annual sugarcane agricultural waste burning emissions (million tonnes of CO₂e). Orange line: São Paulo state. Gray line: all other Brazilian states combined. Source: SEEG (Sistema de Estimativas de Emissões e Remoções de Gases de Efeito Estufa), *Queima de Resíduos Agrícolas* series. São Paulo emissions declined sharply after 2010, coinciding with the phased implementation of the Green Ethanol Protocol (GEP, 2007), which mandated the progressive elimination of pre-harvest burning. Other states show no comparable trend, providing a natural comparison group.

Figure A-5: Distribution of Municipalities by Share of Agricultural Land Allocated to Sugarcane in São Paulo (2002–2010)



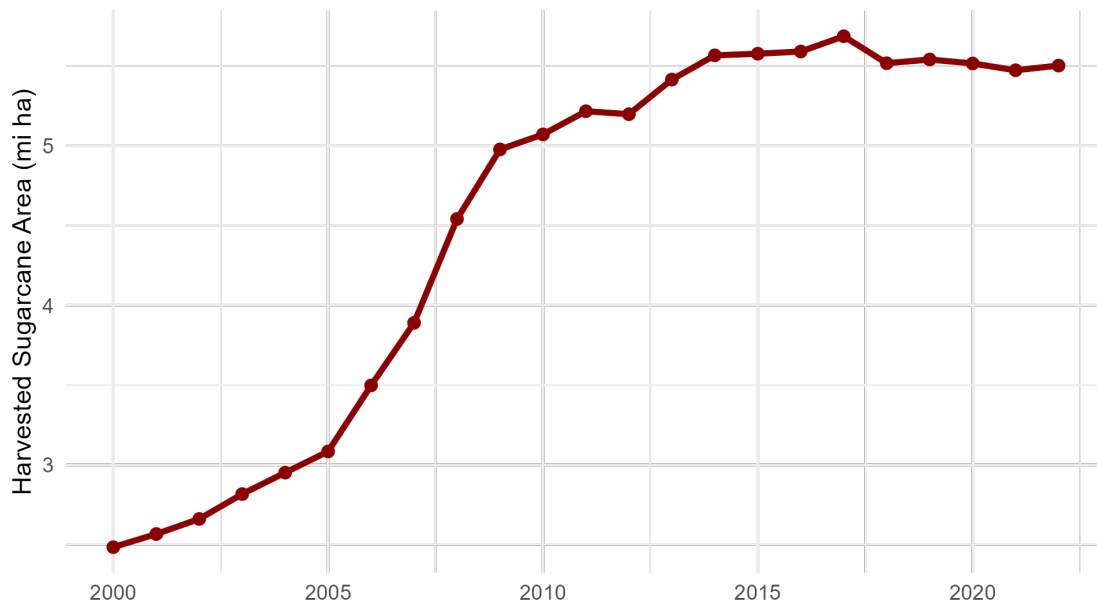
Data from IBGE-Sidra, Pesquisa Agrícola Municipal ([IBGE, 2024](#)). Each bin shows the number of municipalities whose share of total agricultural area devoted to sugarcane falls within the indicated interval. Shares are computed as the ratio of harvested sugarcane area to total harvested cropland area at the municipality level. The last category (>70%) pools all municipalities above that threshold.

Figure A-6: Distribution of Municipalities by Share of Agricultural Land Allocated to Sugarcane in São Paulo (2002–2010)



Data from IBGE-Sidra, Pesquisa Agrícola Municipal ([IBGE, 2024](#)). Shares are computed as the ratio of harvested sugarcane area to total harvested cropland area at the municipality level.

Figure A-7: Harvested Sugarcane Area in São Paulo



Data from IBGE-Sidra, Pesquisa Agrícola Municipal (IBGE, 2024). The figure depicts the harvested sugarcane area in São Paulo state, measured in million hectares.

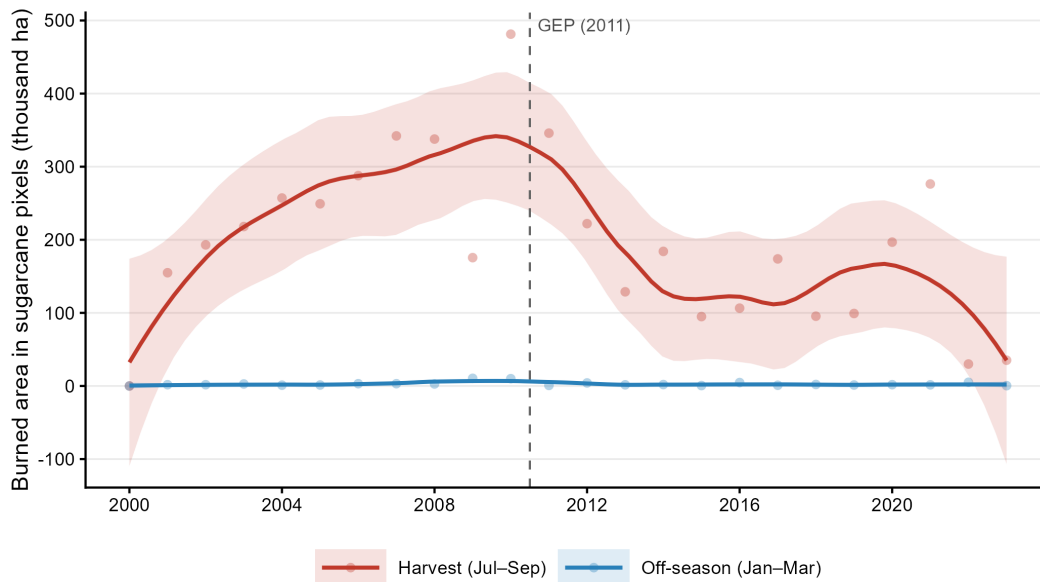
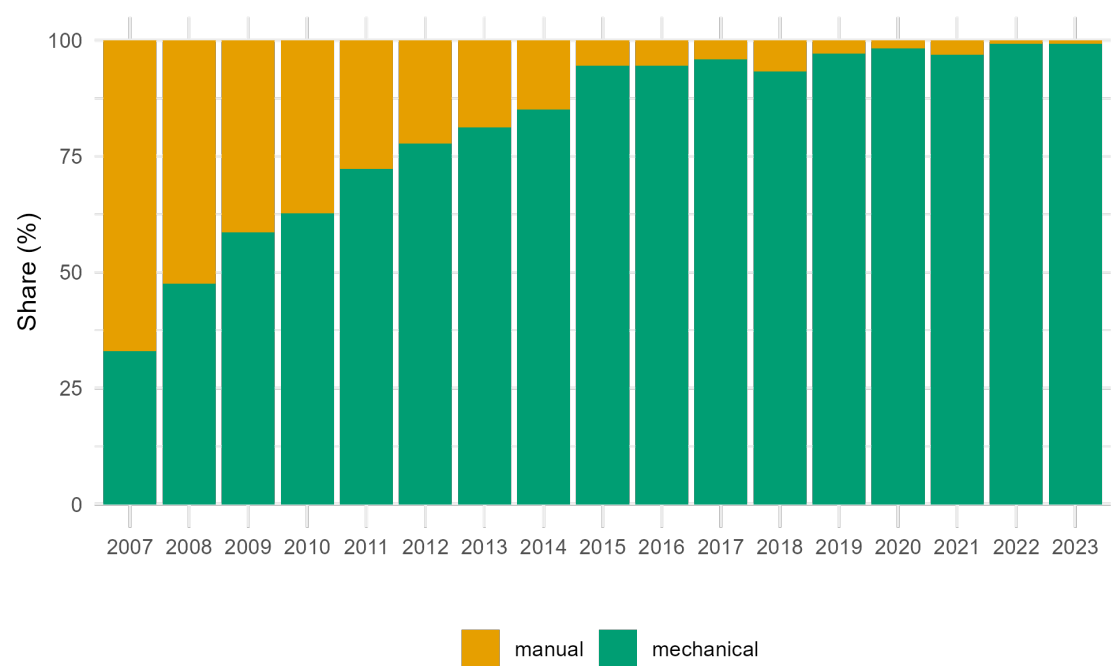


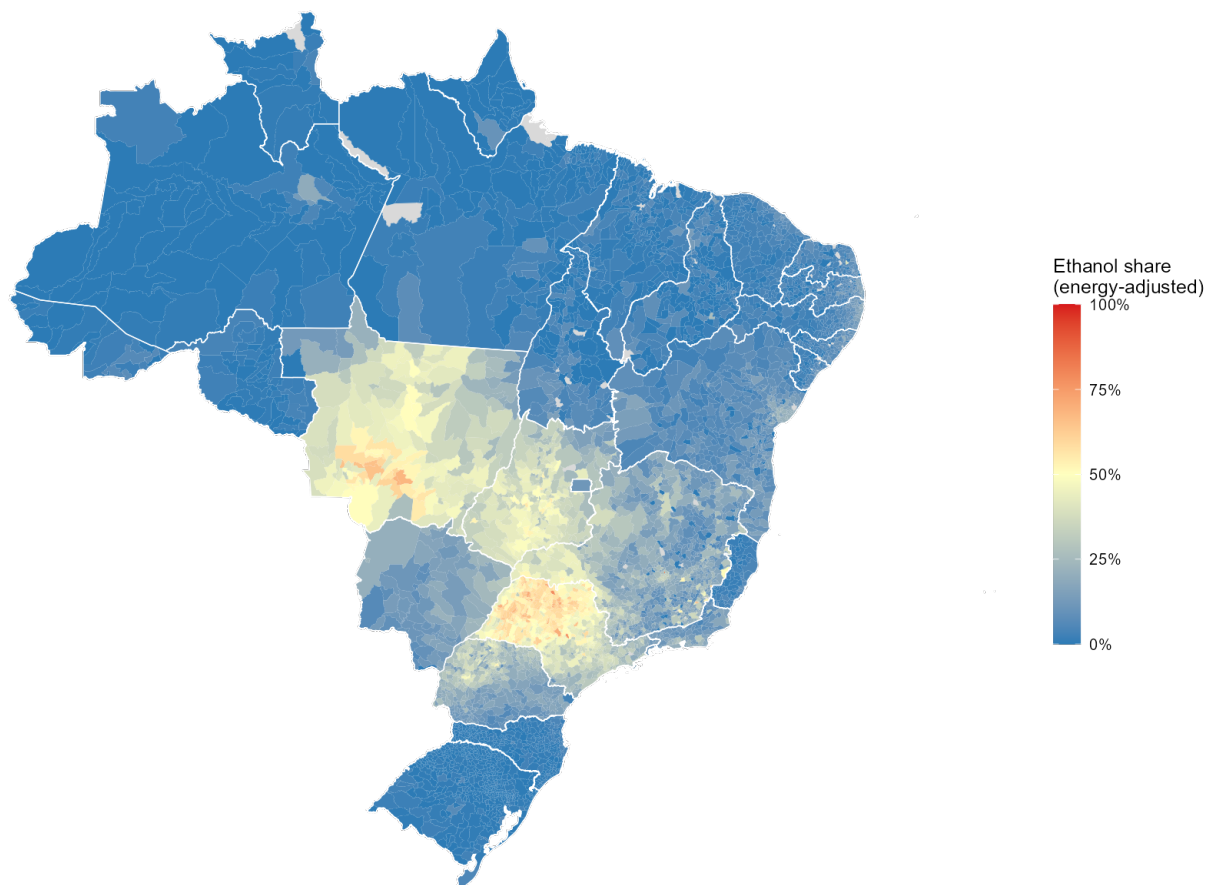
Figure A-8: Sugarcane burned area in São Paulo by season, 2000–2023. Harvest season = July–September; off-season = January–March. Source: MODIS MCD64A1 monthly burned-area product crossed with MapBiomas LULC sugarcane layer (Collection 9), aggregated across all São Paulo municipalities via Google Earth Engine.

Figure A-9: Distribution of Sugarcane Harvesting Methods in São Paulo State (Mechanical vs. Manual)



Data from Conab ([Conab, 2024](#)). The figure illustrates the distribution of harvested sugarcane in São Paulo by harvesting method.

Figure A-10: Energy-Adjusted Hydrated Ethanol Share in the Fuel Mix in Brazil (2020–2024)



Notes: Share defined as $V_{eth} \times (1/1.4) / [V_{eth} \times (1/1.4) + V_{gas}]$, where $1/1.4 \approx 0.714$ is the energy equivalence factor (EPE/ANP standard): one liter of gasoline C contains the same energy as 1.4 litres of hydrated ethanol. Municipal averages computed over 2020–2024. Municipalities with fewer than three years of data are excluded (shown in grey). Source: ANP — *Vendas Anuais de Derivados de Petróleo e Biocombustíveis por Município*.

Figure A-11: Event Study — Respiratory Hospitalizations (Full Sample)

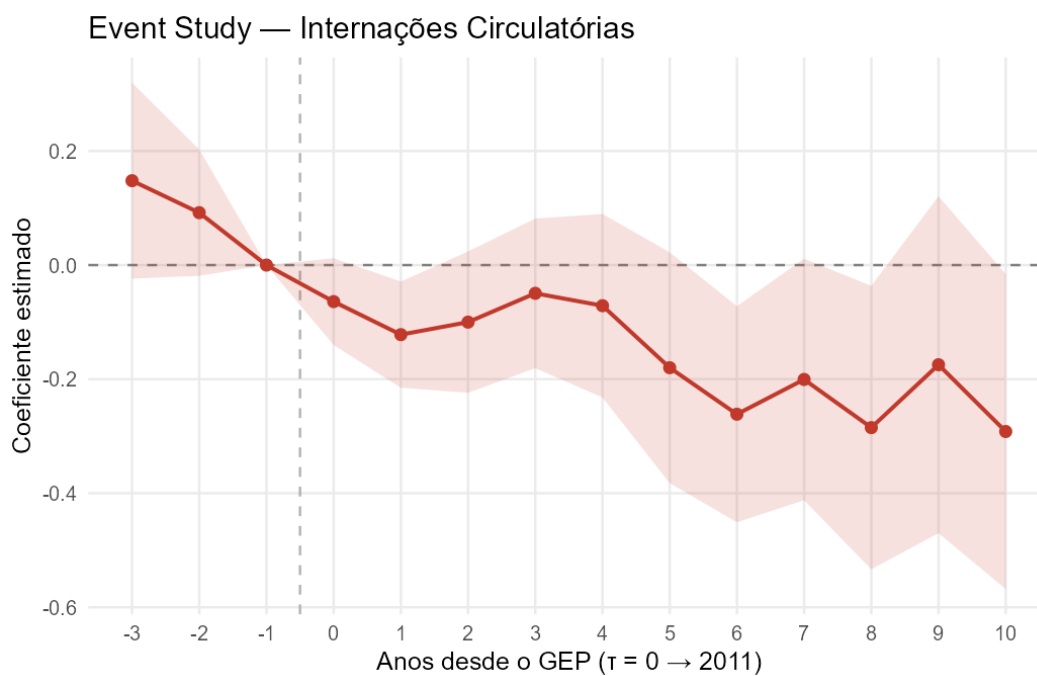
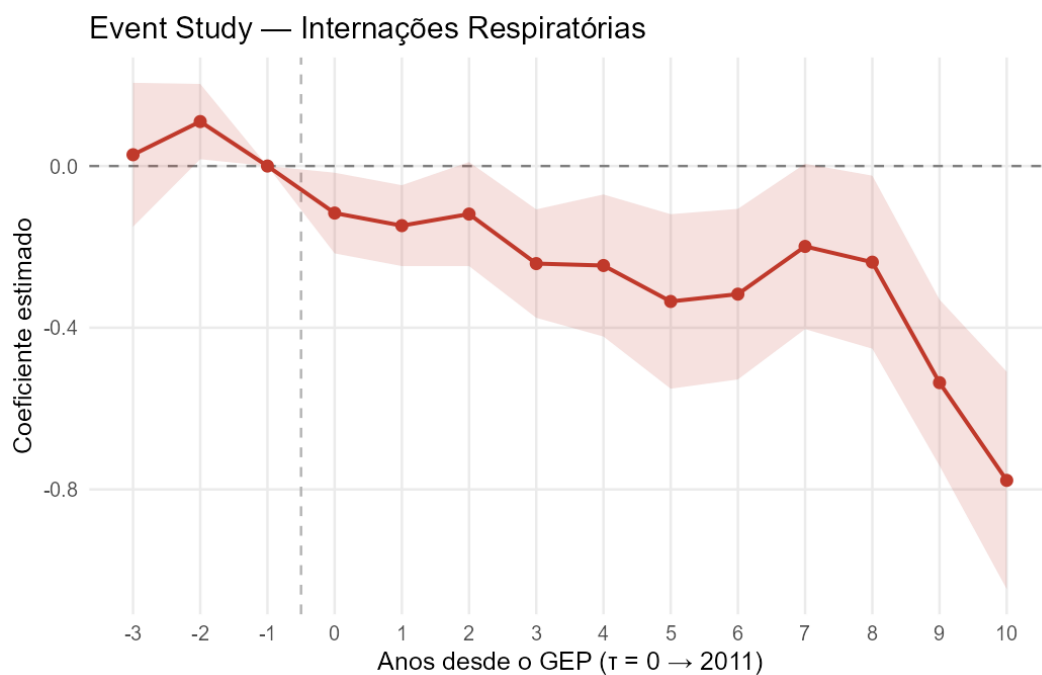
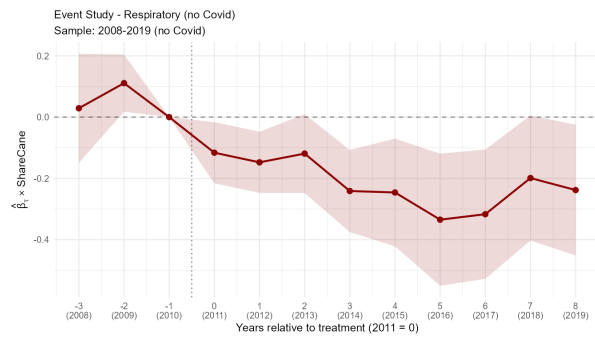
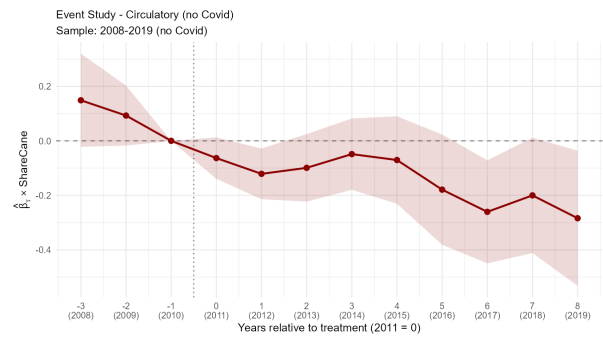


Figure A-12: Event Study — Circulatory Hospitalizations (Full Sample)



(a) Respiratory (Pre-Covid)



(b) Circulatory (Pre-Covid)

Figure A-13: Event Study — Pre-Covid Sample (2008–2019)

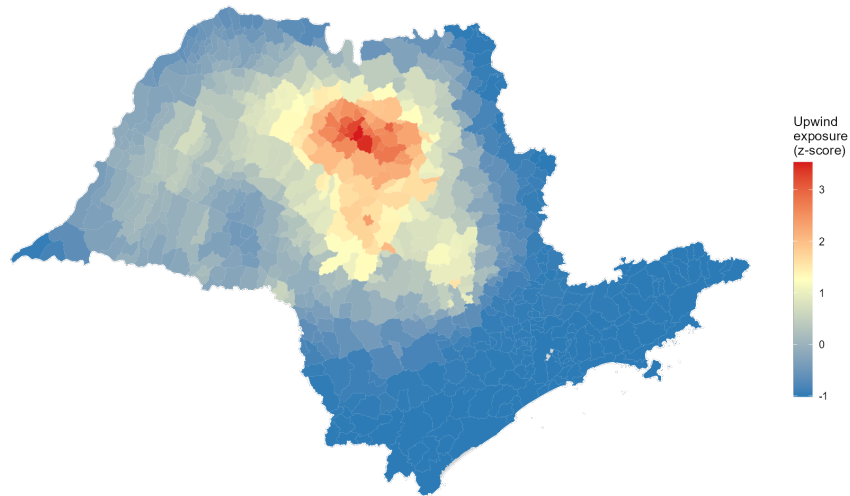
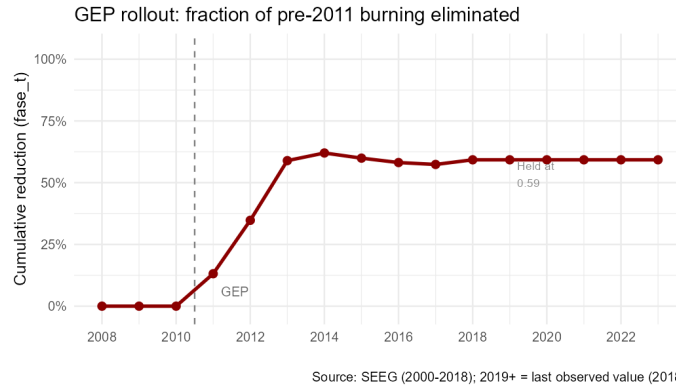


Figure A-14: Upwind Sugarcane Exposure by Municipality (*WindExp*, harvest season average)

Figure A-15: Continuous Treatment Intensity (f_{ase_t}): Fraction of Pre-2011 Sugarcane Burning Emissions Eliminated



f_{ase_t} from SEEG (2000–2018); held at last observed value (0.59) for 2019 onward.

Table A-1: Birth Outcomes — Municipality-Year Aggregate

	$\ln(\text{birthweight})$	$\ln(\text{lbw})$	$\ln(\text{vlbw})$	$\ln(\text{anomalies})$
<i>Panel A — Full sample (2002–2023)</i>				
Exp	−0.002 (0.003)	0.027 (0.023)	−0.067 (0.056)	−0.795** (0.311)
Observations	15,484	15,485	15,485	9,766
<i>Panel B — Pre-Covid (2002–2019)</i>				
Exp	−0.002 (0.004)	0.027 (0.023)	−0.028 (0.054)	−0.789** (0.317)
Observations	12,905	12,906	12,906	8,163
Std. Errors by municipality	✓	✓	✓	✓
Municipality FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Weights	Births	Births	Births	Births

** $p < 0.05$. Municipality-year aggregate outcomes. Outcomes: $\ln(\text{mean birthweight})$, $\ln(\text{share LBW} + 0.001)$, $\ln(\text{share VLBW} + 0.001)$, $\ln(1000 \times \text{anomalies/births} + 0.001)$. Compare with Table 2 for individual-level estimates.

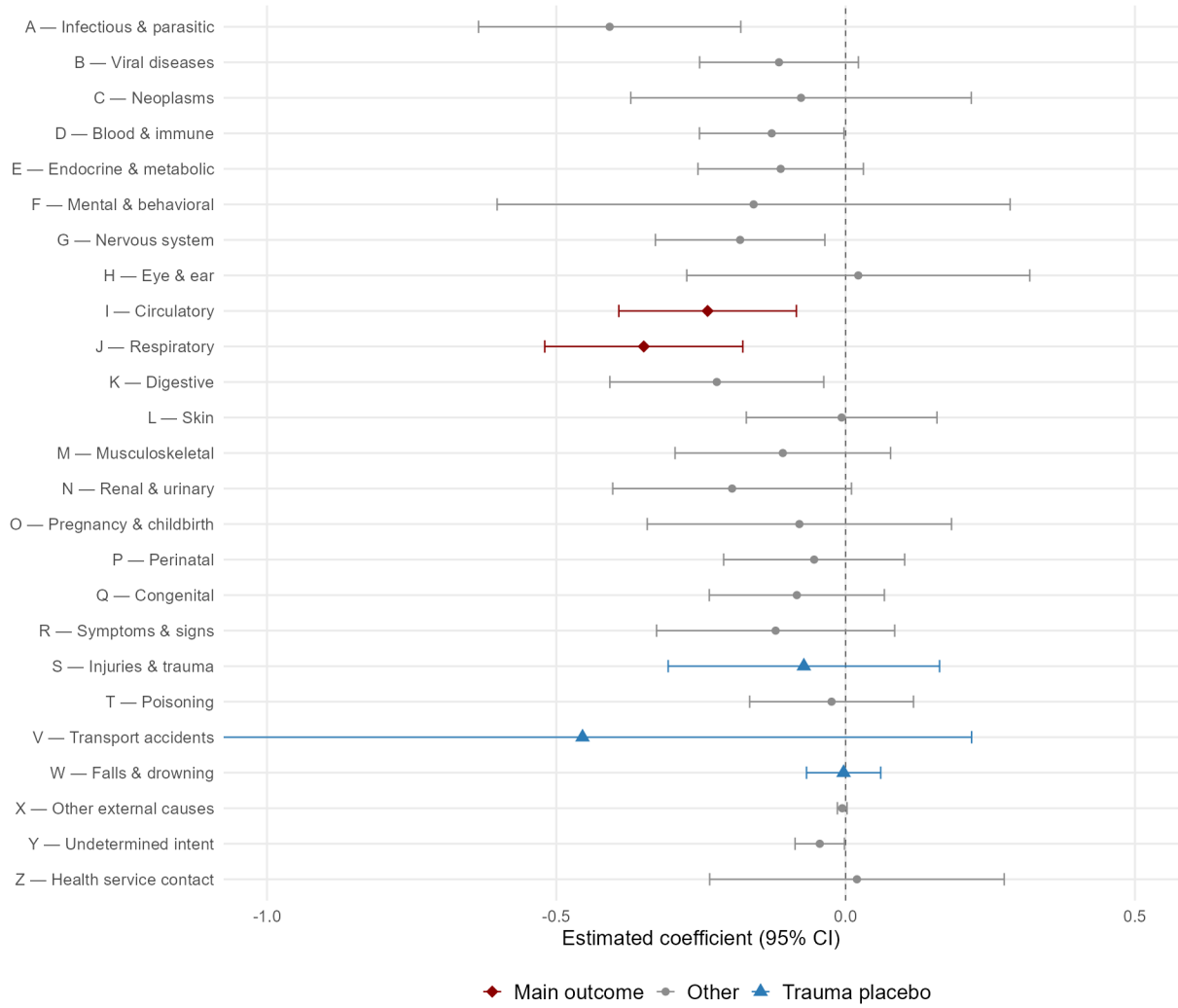


Figure A-16: Effect of sugarcane mechanization across all ICD-10 chapters.

Notes: Each point is $\hat{\beta}^c$ from estimating $\ln(\text{hosp. rate}_{it}^c + 0.001) = \alpha_i + \alpha_t + \beta^c \cdot \text{Exp}_{it} + \varepsilon_{it}$, where $\text{Exp}_{it} = \text{share_cana}_i \times \mathbf{1}(t > 2010) \times \mathbf{1}(\text{share_cana}_i > 5\%)$. Municipality and year fixed effects throughout; standard errors clustered at the municipality level; weighted by population. Panel (B) adds $\ln(\text{GDP per capita})$ as a control. Data: SIH/DataSUS via Base dos Dados, São Paulo, 2008–2021.

Table A-2: Summary Statistics

Variable	N	Mean	SD	Min	Max
<i>Panel A — Treatment variable (municipality level)</i>					
Sugarcane land share (2002–2010)	642	0.150	0.190	0.000	0.886
<i>Panel B — Hospitalization outcomes (municipality × year, 2008–2021)</i>					
Respiratory hosp. per 1,000 pop. (I00–I99)	8,385	7.281	4.939	0.277	45.828
Circulatory hosp. per 1,000 pop. (I00–I99)	8,385	6.470	3.085	0.603	25.760
<i>Panel C — Birth outcomes (municipality × year, 2007–2023)</i>					
Any congenital anomaly (per 1,000 births)	11,882	18.662	91.807	0.000	1,000
Pol3 index: NTD, orofacial & cardiac (per 1,000 births) ^a	9,599,799	3.590	59.809	0.000	1,000
Low birthweight <2,500 g (%)	11,888	9.341	9.510	0.000	100.000
Very low birthweight <1,500 g (%)	11,888	1.550	5.068	0.000	100.000
<i>Panel D — Mortality outcomes (municipality × year, 2002–2023)</i>					
Infant resp. mortality, <5 years (per 1,000 births)	9,029	0.702	3.204	0.000	111.111
Adult resp. mortality (per 100,000 pop.)	9,029	91.935	45.975	0.000	546.448
Adult circ. mortality (per 100,000 pop.)	9,029	199.810	71.132	0.000	650.936

Notes: Unit of observation is the municipality × year for Panels B–D and the municipality for Panel A. Hospitalization rates from SIH/DataSUS (2008–2021); birth outcomes from SINASC (2007–2023); mortality from SIM (2002–2023). São Paulo state municipalities only. Large max values in Panels C and D reflect municipality-year cells with very few births or deaths; all DiD regressions use population or birth weights.

^a Pol3 is an individual-level binary indicator for births presenting at least one of: neural tube defect (Q00–Q07), orofacial cleft (Q35–Q38), or congenital cardiac malformation (Q20–Q26). The N shown is total live births; mean and SD reflect the individual-level binary distribution. All other rows in Panel C are municipality × year cell averages.